Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	TEAM SALAMANCA
Project / Solution Name	Futurique - Your Future. Your Skills. Your Path Domain 1 | SDG 4 and SDG 8
Name (Leader)	CHARUMATHI A/P SATHIA GUNALAN
Email & Phone (Leader)	scharu2772@gmail.com (011-10745483)
University Affiliation(s)	Universiti Teknologi PETRONAS; Universiti Teknologi Malaysia; Asia Pacific University; Xiamen University Malaysia
Live Prototype URL	https://futurique.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Peacedrama-11/Team-Salamanca

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Futurique uses a React 19 and TypeScript frontend built with Vite, Tailwind CSS and Radix UI. The browser sends REST requests to a Node.js 24 / Express 5 API through the /api route. PostgreSQL 16 and Drizzle ORM store public vacancies and collection status. Python 3.13 processes PDF and DOCX CVs using pypdf and python-docx; dictionary rules extract skills for user review. Deterministic matching compares confirmed skills with recognized requirements in cached job descriptions. Python collectors retrieve public vacancies from Ashby, Greenhouse and Lever. The API also retrieves official Malaysian labour statistics from data.gov.my. Learning recommendations use questionnaire rules and link to roadmap.sh. Selected learning paths and confirmed CV skills can persist in browser local storage. Replit supplies the development environment and deployment configuration for the web and API services. API contracts use OpenAPI, Orval and Zod. No LLM, trained model, RunPod endpoint or GPU inference is integrated in this version.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React / React DOM 19.1.0	Web interface and rendering	MIT
TypeScript 5.9.3	Typed application and shared API code	Apache-2.0
Node.js 24 / Python 3.13	API runtime; collection and CV processing	Node.js: MIT and bundled notices; Python: PSF
Vite 7.3.7 / esbuild 0.28.2	Frontend development and application builds	MIT
Tailwind CSS 4.3.3 / Radix UI 1.x-2.x	Styling and accessible interface primitives	MIT

Name & Version	Purpose in Project	License
Wouter 3.13.0	Client-side routing	Unlicense
TanStack React Query 5.104.1 / Recharts 2.15.4	API request state and workforce charts	MIT
Framer Motion 12.43.0 / React Hook Form 7.90.0	Interface animation and form helpers	MIT
Lucide React 0.545.0	Interface icons	ISC
Express 5.3.0 / Multer 2.5.0	REST API and bounded in-memory file uploads	MIT
CORS 2.8.6 / cookie-parser 1.4.7 / Pino 9.14.0	HTTP middleware and request logging	MIT
PostgreSQL 16 / pg 8.23.1	Public vacancy cache and database connection	PostgreSQL License / MIT
Drizzle ORM 0.45.4	Typed database access and schemas	Apache-2.0
Drizzle Kit 0.31.11 / Zod 3.25.76 and 4.6.5	Development schema tooling and validation	MIT
Orval 8.41.0 / OpenAPI	Generated React API client and validation schemas	Orval: MIT; OpenAPI specification
pypdf 6.21.2	Extract text from readable PDF CVs	BSD-3-Clause
python-docx 1.2.0	Extract text from DOCX CVs	MIT
mherzog4/job-boards revision da7885cff552	Vendored scanner, source adapters and seed boards	MIT; upstream license retained
Replit Vite plugins 0.x / connectors-sdk 0.4.3	Replit preview/development support; SDK dependency present	Provider/package terms; license confirmation to be added
pnpm / uv (tool versions to be added)	Workspace packages and locked Python dependencies	pnpm: MIT; uv: MIT OR Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
data.gov.my / Department of Statistics Malaysia	Monthly labour-force data: lfs_month	Free public API; no key
Ashby, Greenhouse and Lever public job-board APIs	Vacancies and available descriptions from configured company boards	Public endpoints; no paid account/key used
Google Fonts	Browser downloads DM Sans and Space Grotesk fonts	Free public font service
Internet Archive CDX (optional discovery script)	Optional operator-run job-board discovery; outside normal app requests	Public endpoint; optional use

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Project development; static frontend and autoscaled API deployment configuration	Plan / region / credit source: To be added
PostgreSQL database	Shared public vacancy cache through DATABASE_URL	Provider / hosting plan / region: To be added
GitHub	Public source repository and version control	Public repository; account plan / region not specified

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None	Not applicable	No pre-trained, fine-tuned or self-trained model is executed. Extraction and recommendations use deterministic rules.	Not applicable

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Monthly Principal Labour Force Statistics	DOSM OpenDOSM lfs_month	Public; running API and local reference CSV; aggregate statistics	CC BY 4.0	N
Annual Salaries and Wages by Occupation and Sex	DOSM OpenDOSM wages_occupation	Public; reference CSV; aggregate statistics; not connected in current app	CC BY 4.0	N
Annual Salaries and Wages by Industry and Sex	DOSM OpenDOSM wages_industry	Public; reference CSV; aggregate statistics; not connected in current app	CC BY 4.0	N
Quarterly Skills-Related Underemployment by Age	DOSM OpenDOSM lfs_qtr_sru_age	Public; reference CSV; aggregate statistics; not connected in current app	CC BY 4.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Employment by MSIC Sector and Sex	DOSM OpenDOSM employment_sector	Public; reference CSV; aggregate statistics; not connected in current app	CC BY 4.0	N
Public vacancies and available descriptions	Ashby / Greenhouse / Lever company boards	Public; cached by the app; may contain public contact details	Provider / employer terms; no blanket open-data license asserted	Y
Job-board seed list	mherzog4/job-boards boards.seed.json	Public; vendored board identifiers	MIT	N
Demo profiles, learning rules, scenarios and CV fixtures	Project source and generated synthetic test documents	Synthetic; fictional examples and regression evaluation	Project repository license; third-party notices retained	N
User-supplied CV / confirmed skills	User input through the CV workspace	Private input; real CVs may contain personal data; no model training	User-provided material; no redistribution license asserted	Y

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✓ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent (version not recorded)	Application scaffolding and iterative feature/code generation; Replit Agent commits are present in history	Major
Gemini	To prompt appropriate wording for proposal and slides	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-assisted code and documentation were checked against the repository, API contracts and official dataset metadata. Existing regression tests were executed: 18 frontend logic tests and 6 vacancy collector tests passed. Documentation was corrected where older notes contradicted implemented CV processing. Matching exposes evidence and unknown requirements, and scenarios carry explicit non-prediction labels. A complete application build and live deployment verification remain pending. The team leader must review this declaration, confirm all team tool usage and accept responsibility for the final submission.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✓ Yes ☐ No To be added: confirm official sponsor allocation.

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Development platform; sponsorship / credits status to be added	Used to develop the project; repository contains Replit runtime, preview and deployment configurations
RunPod	None used in the current implementation	No handler, endpoint or GPU workload is integrated

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit supported development and provides deployment configuration, but its platform can be replaced by another host supporting a static React frontend, Node.js, Python and PostgreSQL. The core extraction, matching and learning rules do not depend on a proprietary AI API. No RunPod workload is currently integrated. Whether Replit access or any other resource was supplied through official sponsor credits remains to be confirmed by the team.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demo can accept an optional email, skills, experience and a PDF/DOCX CV. CV bytes are sent to our API for in-memory parsing, bounded to 8 MB, and the upload buffer is cleared after processing; raw files are not saved in PostgreSQL. Confirmed skills and learning selections persist in browser local storage until removed, scoped by demo email rather than secure authentication. Other dashboard and HR records use demo state. Public vacancies and collection metadata persist in PostgreSQL. A supplied email may be sent as scraper contact to job providers; blank input uses temp@gmail.com. Request logs omit bodies and contact addresses, while short-lived IP counters limit extraction requests. No CV content is sent to an AI model. Google Fonts receives normal browser requests. Use synthetic CVs for demonstrations; cloud-provider log retention details are to be added.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

No deployed LLM or trained model is present. Dictionary extraction can miss skills, and rule-based matching depends on incomplete job descriptions. Overlap scores do not measure proficiency or hiring probability; missing evidence is shown as unknown and users review extracted skills. Scanned PDFs lack OCR support. Job-board coverage is partial, may be stale and is not representative of the Malaysian labour market; source links, timestamps, collection status and cached snapshots expose these limits. Parsing has upload, time and concurrency limits with manual entry as a fallback. Career Crash Test scenarios, HR ratings and SDG links are illustrative. Email-scoped browser storage is not secure authentication or cross-device persistence. Full deployment verification and a final license/privacy review are still required.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✓ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✓ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✓ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✓ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✓ Any use of official SDC sponsor resources has been declared in Section 5.
- ✓ The prototype described here is accessible to judges through the links provided in Section 1.
- ✓ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature:



Date: 25/10/2026